

# Abdelrahman Taeha

916-385-3911 | [abdeltaehass@gmail.com](mailto:abdeltaehass@gmail.com) | [abdeltaehass.github.io](https://github.com/abdeltaehass) | [linkedin.com/in/abdel-rahman-taeha](https://linkedin.com/in/abdel-rahman-taeha) | [github.com/abdeltaehass](https://github.com/abdeltaehass)

## Education

### Georgia Institute of Technology

August 2026 – May 2028

M.S. Computer Science (Artificial Intelligence Concentration)

### California State University, Sacramento

December 2025

B.S. Computer Science

## Professional Experience

### AI Research Engineer | Medical AI Research

August 2025 – Present

- Co-authored [SpineFairBench](#), a counterfactual vision-language model (VLM) benchmark auditing demographic bias in spinal radiology report generation.
  - Evaluated 7,996 paired radiographs across a frozen 9-model panel with radiologist validation, surfacing demographic recommendation drift across all evaluated models.
  - Trained a neutral baseline model and built data collection pipelines for 2 major open-access clinical repositories to isolate demographic bias in VLM report generation.
- Co-developed MedInsider, a FHIR-style benchmark measuring whether medical LLM agents preserve decision integrity when exposed to institutional and financial pressure.
  - Constructed 840 paired-twin scenarios across 5 regulatory failure families and 8 incentive conditions to isolate when and why agents deviate from compliant behavior.
  - Engineered simulated EHR tooling with deterministic action-log scoring, run auditing, and reproducibility manifests to evaluate tool-use agents across 9 medical vision-language models.

### Software Engineer | DHCS

May 2025 – Present

- Automated monitoring and reporting pipelines across enterprise systems, reducing manual observability overhead by 40% and cutting mean time to detection (MTTD) for production incidents in half.
  - Built backend integrations and validation workflows that surfaced anomalies early, accelerating root-cause analysis by 35%.
- Established internal dashboards visualizing agent-driven operational metrics, reducing incident resolution time by 30%; improved real-time alerting accuracy by 20% through log-processing pipelines.

### Software Engineering Intern (Machine Learning) | District Hut

May 2024 – July 2025

- Scaled distributed data pipelines processing 3 TB/week across cloud infrastructure, improving throughput and system reliability in a high-volume production environment.
  - Revamped SQL data-integrity checks, preventing 5 critical production data-corruption issues and automating 75% of validation tasks with 100% test coverage.
- Collaborated with 3 engineers to automate model retraining agents, reducing collective manual effort by 6 hours per week via autonomous performance metrics.

## Project Highlights

### R.A.D.A.R — Vision-Language Radiology Report Generator — [github.com/abdeltaehass/RADAR](https://github.com/abdeltaehass/RADAR)

- Built and deployed an end-to-end vision-language system that generates chest-X-ray reports from images, parameter-efficient fine-tuning BLIP-2 with LoRA (training 0.33% of 3.76B params) on 3,337 patient-split IU X-Ray studies.
- Quantified and mitigated hallucination (the core VLM failure mode) with reference-grounded analysis across decoding strategies, cutting the hallucination rate from 40% to 10% via confidence-threshold abstention.
- Engineered a clinical-term overlap metric exposing a fluent baseline scoring 44 ROUGE-L but 0 clinical-F1; shipped an interactive Gradio demo with per-sentence Grad-CAM grounding on a quantized Hugging Face Space.

### S.A.I.N.T. — Network Intrusion Detection System — [github.com/abdeltaehass/S.A.I.N.T](https://github.com/abdeltaehass/S.A.I.N.T)

- Partnered with 2 peers to build an AI agent reasoning engine for threat detection, reaching 89% accuracy on NSL-KDD (125K+ samples) via feature engineering across 41 traffic-flow features.
- Designed a low-latency inference agent using Flask + Redis + Plotly, processing 50–100 network connections/sec with an interactive agent-in-the-loop dashboard for analyst review.

### PingVault — Network Observability Agent — [github.com/abdeltaehass/PingVault](https://github.com/abdeltaehass/PingVault)

- Created a production-grade network monitoring tool spanning 3 protocols (ICMP, SNMP, TCP/UDP) with agent-based metrics collection, Alembic migrations, and a real-time HTML observability dashboard mirroring enterprise SRE tooling.

## Skills

**Languages:** Python, Java, C/C++, SQL, JavaScript, Swift, MATLAB, R

**ML & AI:** PyTorch, TensorFlow, scikit-learn, pandas, NumPy, LLM Orchestration (LangChain/CrewAI), RAG, Autonomous Agents, Prompt Engineering, Model Evaluation

**Frameworks & Cloud:** React.js, Node.js, Spring Boot, Flask, FastAPI, AWS, GCP, Docker, Git, GitHub Actions, CI/CD

**Research & Domains:** VLM Evaluation, Counterfactual Benchmarking, Medical Imaging, Bias Auditing, SIEM